

Rosen — §1.5: Nested Quantifiers Selected Exercises

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Semester 2, 2026

Exercise 1

Translate these statements into English, where the domain for each variable consists of all real numbers.

- (a) $\forall x \exists y (x < y)$
- (b) $\forall x \forall y (((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0))$
- (c) $\forall x \forall y \exists z (xy = z)$

Solution. **(a)** $\forall x \exists y (x < y)$

For all x , there exists y such that $x < y$. (For every real number, there is a larger real number.)

(b) $\forall x \forall y ((x \geq 0 \wedge y \geq 0) \Rightarrow xy \geq 0)$

For all real numbers x and y , if x and y are both nonnegative, then their product xy is nonnegative. (The product of two nonnegative real numbers is nonnegative.)

(c) $\forall x \forall y \exists z (xy = z)$

For all x and y , there exists z such that $xy = z$. (Every pair of real numbers has a product.) ■

Exercise 7

Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.

- (a) $\neg T(\text{Abdallah Hussein}, \text{Japanese})$
- (b) $\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$
- (c) $\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$
- (d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$
- (e) $\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$
- (f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$

Solution. Let $T(x, y)$ mean “student x likes cuisine y .”

(a) $\neg T(\text{Abdallah Hussein}, \text{Japanese})$

Abdallah Hussein does not like Japanese cuisine.

(b) $\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$

There is a student who likes Korean cuisine, and every student likes Mexican cuisine.

(c) $\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$

There is a cuisine that Monique Arsenault or Jay Johnson likes.

(d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$

For any two different students, there is some cuisine that they don’t both like.

(e) $\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$

There exist two students x and z such that, for every cuisine y , x likes y if and only if z likes y . (There are two students with exactly the same cuisine preferences.)

(f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$

For all students x and z , there is some cuisine y such that x likes y if and only if z likes y . (This is trivially true for every pair of students, e.g. any cuisine both like or both dislike.) ■

Exercise 9

Let $L(x, y)$ be the statement “ x loves y ,” where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.

(a) Everybody loves Jerry.

(b) Everybody loves somebody.

(c) There is somebody whom everybody loves.

(d) Nobody loves everybody.

(e) There is somebody whom Lydia does not love.

(f) There is somebody whom no one loves.

(g) There is exactly one person whom everybody loves.

(h) There are exactly two people whom Lynn loves.

(i) Everyone loves himself or herself.

(j) There is someone who loves no one besides himself or herself.

Solution. Let $L(x, y)$ mean “ x loves y ,” with domain all people.

(a)

$$\forall x L(x, \text{Jerry})$$

(b)

$$\forall x \exists y L(x, y)$$

(c)

$$\exists y \forall x L(x, y)$$

(d)

$$\neg \exists x \forall y L(x, y) \equiv \forall x \exists y \neg L(x, y)$$

(e)

$$\exists x \neg L(\text{Lydia}, x)$$

(f)

$$\exists x \forall y \neg L(y, x)$$

(g)

At least one:

$$\exists x \forall y L(y, x)$$

At most one:

$$\forall x_1 \forall x_2 \left((\forall y L(y, x_1) \wedge \forall y L(y, x_2)) \Rightarrow x_1 = x_2 \right)$$

Exactly one (conjunction of the above):

$$\exists x \forall y L(y, x) \wedge \forall x_1 \forall x_2 \left((\forall y L(y, x_1) \wedge \forall y L(y, x_2)) \Rightarrow x_1 = x_2 \right)$$

Using the uniqueness template $\exists! x P(x) \equiv \exists x (P(x) \wedge \forall z (P(z) \Rightarrow z = x))$ with $P(x) \equiv \forall y L(y, x)$, this simplifies to:

$$\exists x \left(\forall y L(y, x) \wedge \forall z (\forall y L(y, z) \Rightarrow z = x) \right)$$

(h)

At least two:

$$\exists x_1 \exists x_2 ((x_1 \neq x_2) \wedge L(\text{Lynn}, x_1) \wedge L(\text{Lynn}, x_2))$$

At most two:

$$\forall x_1 \forall x_2 \forall x_3 \left((L(\text{Lynn}, x_1) \wedge L(\text{Lynn}, x_2) \wedge L(\text{Lynn}, x_3)) \Rightarrow (x_1 = x_2 \vee x_1 = x_3 \vee x_2 = x_3) \right)$$

Exactly two (combining at least two with at most two, reusing x_1, x_2):

$$\begin{aligned} & \exists x_1 \exists x_2 \left((x_1 \neq x_2) \wedge L(\text{Lynn}, x_1) \wedge L(\text{Lynn}, x_2) \right. \\ & \left. \wedge \forall x_3 (L(\text{Lynn}, x_1) \wedge L(\text{Lynn}, x_2) \wedge L(\text{Lynn}, x_3) \Rightarrow (x_1 = x_2 \vee x_1 = x_3 \vee x_2 = x_3)) \right) \end{aligned}$$

(i)

$$\forall x L(x, x)$$

(j)

$$\exists x \forall y (x \neq y \Rightarrow \neg L(x, y))$$

■

Exercise 13

Let $M(x, y)$ be “ x has sent y an e-mail message” and $T(x, y)$ be “ x has telephoned y ,” where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all e-mail messages that were sent are received, which is not the way things often work.)

- (a) Chou has never sent an e-mail message to Koko.
- (b) Arlene has never sent an e-mail message to or telephoned Sarah.
- (c) José has never received an e-mail message from Deborah.
- (d) Every student in your class has sent an e-mail message to Ken.
- (e) No one in your class has telephoned Nina.
- (f) Everyone in your class has either telephoned Avi or sent him an e-mail message.
- (g) There is a student in your class who has sent everyone else in your class an e-mail message.
- (h) There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class.
- (i) There are two different students in your class who have sent each other e-mail messages.
- (j) There is a student who has sent himself or herself an e-mail message.
- (k) There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.
- (l) Every student in the class has either received an e-mail message or received a telephone call from another student in the class.
- (m) There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student.
- (n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class.

Solution. Let $M(x, y)$ mean “ x has sent y an e-mail message” and $T(x, y)$ mean “ x has telephoned y .”

(a)

$$\neg M(\text{Chou}, \text{Koko})$$

(b)

$$\neg (M(\text{Arlene}, \text{Sarah}) \vee T(\text{Arlene}, \text{Sarah}))$$

(c)

$$\neg M(\text{Deborah}, \text{José})$$

(d)

$$\forall x M(x, \text{Ken})$$

(e)

$$\forall x \neg T(x, \text{Nina})$$

(f)

$$\forall x (T(x, \text{Avi}) \vee M(x, \text{Avi}))$$

(g)

$$\exists x \forall y (x \neq y \Rightarrow M(x, y))$$

(h)

$$\exists x \forall y (x \neq y \Rightarrow (M(x, y) \vee T(x, y)))$$

(i)

$$\exists x \exists y (x \neq y \wedge M(x, y) \wedge M(y, x))$$

(j)

$$\exists x M(x, x)$$

(k)

$$\exists x \forall y ((x \neq y) \Rightarrow (\neg M(y, x) \wedge \neg T(y, x)))$$

(l)

$$\forall x \exists y (x \neq y \wedge (T(y, x) \vee M(y, x)))$$

(m)

$$\exists x_1 \exists x_2 (M(x_1, x_2) \wedge T(x_2, x_1))$$

(n)

$$\begin{aligned} & \exists x_1 \exists x_2 \forall y ((x_1 \neq x_2 \wedge x_1 \neq y \wedge x_2 \neq y) \\ & \Rightarrow (M(x_1, y) \vee M(x_2, y) \vee T(x_1, y) \vee T(x_2, y))) \end{aligned}$$

■

Exercise 17

Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.

- (a) Every user has access to exactly one mailbox.
- (b) There is a process that continues to run during all error conditions only if the kernel is working correctly.
- (c) All users on the campus network can access all websites whose url has a .edu extension.
- (d) There are exactly two systems that monitor every remote server.

Solution.

(a) Let $A(x, y)$: “user x has access to mailbox y .”

$$\forall x \exists y (A(x, y) \wedge \forall z (A(x, z) \Rightarrow y = z))$$

(b) Let $C(x, y)$: “process x continues to run during error condition y ”; K : “the kernel is working correctly.”

$$\exists x ((\forall y C(x, y)) \Rightarrow K)$$

(c) Let $A(x, y)$: “user x on the campus can access website y ”; $U(y)$: “website y has a .edu extension.”

$$\forall x \forall y (U(y) \Rightarrow A(x, y))$$

■

Exercise 21

Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that every positive integer is the sum of the squares of four integers.

Solution.

$$\forall x (x > 0 \Rightarrow \exists y_1 \exists y_2 \exists y_3 \exists y_4 (x = y_1^2 + y_2^2 + y_3^2 + y_4^2))$$

■

Exercise 23

Express each of these mathematical statements using predicates, quantifiers, logical connectives, and mathematical operators.

- (a) The product of two negative real numbers is positive.
- (b) The difference of a real number and itself is zero.
- (c) Every positive real number has exactly two square roots.
- (d) A negative real number does not have a square root that is a real number.

Exercise 31

Express the negations of each of these statements so that all negation symbols immediately precede predicates.

- (a) $\forall x \exists y \forall z T(x, y, z)$
- (b) $\forall x \exists y P(x, y) \vee \forall x \exists y Q(x, y)$
- (c) $\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$
- (d) $\forall x \exists y (P(x, y) \rightarrow Q(x, y))$

Exercise 37

Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)

- (a) Every student in this class has taken exactly two mathematics classes at this school.
- (b) Someone has visited every country in the world except Libya.
- (c) No one has climbed every mountain in the Himalayas.
- (d) Every movie actor has either been in a movie with Kevin Bacon or has been in a movie with someone who has been in a movie with Kevin Bacon.

Exercise 45

Determine the truth value of the statement $\forall x \exists y (xy = 1)$ if the domain for the variables consists of

- (a) the nonzero real numbers.
- (b) the nonzero integers.
- (c) the positive real numbers.

Exercise 47

Show that the two statements $\neg \exists x \forall y P(x, y)$ and $\forall x \exists y \neg P(x, y)$, where both quantifiers over the first variable in $P(x, y)$ have the same domain, and both quantifiers over the second variable in $P(x, y)$ have the same domain, are logically equivalent.

Exercise 49

- (a) Show that $\forall x P(x) \wedge \exists x Q(x)$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same nonempty domain.
- (b) Show that $\forall x P(x) \vee \exists x Q(x)$ is equivalent to $\forall x \exists y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain.

Exercise 51

A statement is in *prenex normal form* (PNF) if and only if it is of the form

$$Q_1 x_1 Q_2 x_2 \cdots Q_k x_k P(x_1, x_2, \dots, x_k),$$

where each Q_i , $i = 1, 2, \dots, k$, is either the existential quantifier or the universal quantifier, and $P(x_1, \dots, x_k)$ is a predicate involving no quantifiers. For example, $\exists x \forall y (P(x, y) \wedge Q(y))$ is in prenex normal form, whereas $\exists x P(x) \vee \forall x Q(x)$ is not (because the quantifiers do not all occur first).

Every statement formed from propositional variables, predicates, T , and F using logical connectives and quantifiers is equivalent to a statement in prenex normal form.

Show how to transform an arbitrary statement to a statement in prenex normal form that is equivalent to the given statement.

Note: A formal solution of this exercise requires use of structural induction, covered in Section 5.3.